



**GIRRAWEE HIGH SCHOOL
YEAR 11 YEARLY EXAMINATION**

2008

MATHEMATICS

2 UNIT

Time allowed – Two Hours

(plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every questions.
- Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new page.
- Write your name on each piece of paper.

Question 1 (16 marks)

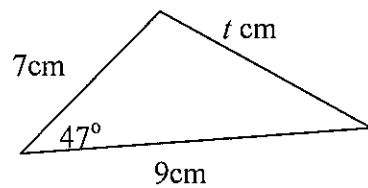
- (a) Find, correct to two decimal places: $\sqrt[3]{\frac{64.9 \times 13.4}{862 \times 3.69}}$ **1**
- (b) Express $\frac{x-6}{x^2+4} + \frac{x+2}{x^2-4}$ as a single fraction and simplify. **3**
- (c) Factorise fully:
- (i) $4x^2 + 16x - 9$ (ii) $27 - y^3$ **3**
- (d) Show that $\frac{1}{3-\sqrt{2}} + \frac{1}{3+\sqrt{2}}$ is a rational number. **2**
- (e) Solve the following pair of simultaneous equations. **3**
- $$\begin{aligned} 6x + 3y &= 0 \\ 2x - 5y &= 6 \end{aligned}$$
- (f) Solve $|3x-1| > 5$ and graph the solution on a number line. **4**

Question 2 (21 marks)

(a) Find the exact value of:

(i) $\cos 150^\circ$ (ii) $\tan 300^\circ$ 4

(b)



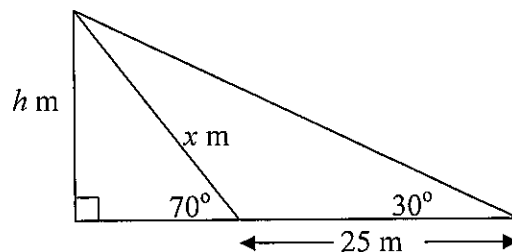
Find:

- (i) the value of t , correct to one decimal place. 2
 (ii) area of the triangle, correct to one decimal place. 2

(c) Solve the following for $0^\circ < \theta < 360^\circ$.

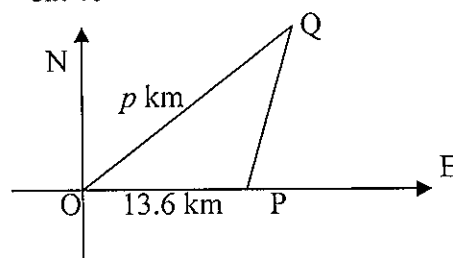
(i) $2 \cos \theta + 1 = 0$ (ii) $\tan^2 \theta = 3$ 5

(d)



Prove that $x = \frac{25 \sin 30}{\sin 40}$. Hence find the value of h , correct to one decimal place. 3

(e)



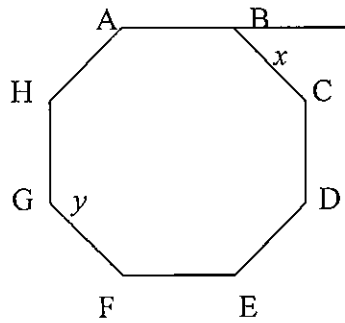
P is 13.6 km due east of O. The bearings of Q from O and P are 053° and 027° respectively. Find p , correct to one decimal place. 3

(f) Simplify $\frac{1 - \cos^2 \theta}{1 - \sin^2 \theta}$ 2

Question 3 (11 marks)

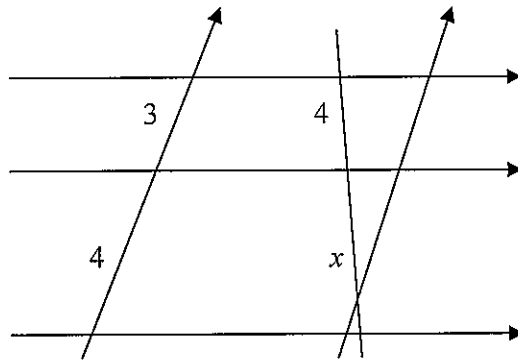
- (a) ABCDEFGH is a regular octagon.
Find the values of x and y , giving reasons.

4



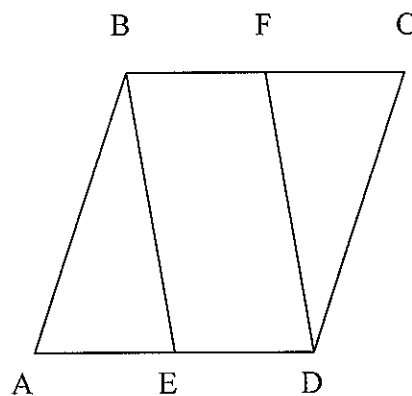
- (b) Find the value of x , giving reasons.

3



- (c) ABCD is a rhombus. E and F are midpoints of AD and BC respectively.
Prove that EBFD is a parallelogram.

4



Question 4 (24 marks)

- (a) Sketch the following functions, showing any intercepts with coordinate axes and stating their domain and range.

(i) $y = x^2 - 4$ 4

(ii) $y = \frac{2}{x}$ 3

(iii) $y = \sqrt{5 - x}$ 4

- (b) Show that the following function is odd:

$$f(x) = x^3 + 4x$$

Hence, or otherwise, evaluate $f(2) + f(-2)$ 3

- (c) Shade the region corresponding to each of the following inequalities on separate number planes.

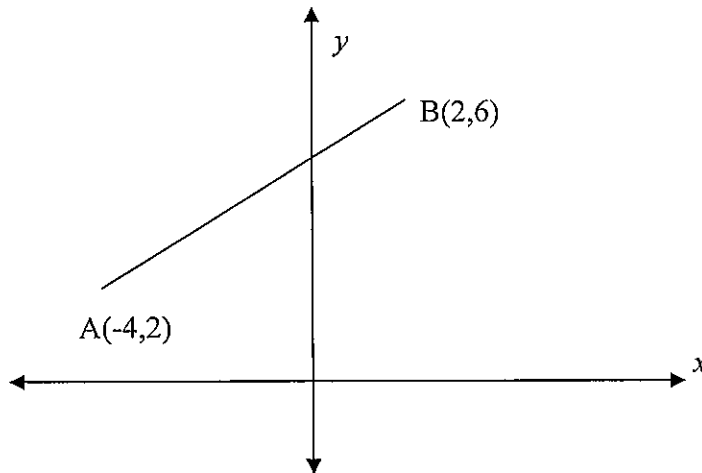
(i) $y > x^2 - 2$ 3

(ii) $xy \leq 1$ 4

- (d) Sketch $y = \begin{cases} 2 - x^2 & x < 0 \\ -x & x \geq 0 \end{cases}$ 3

Question 5 (20 marks)

- (a) The diagram shows two points A(-4,2) and B(2,6) on the number plane.



Copy the diagram onto your answer sheet:

- (i) Find the coordinates of the midpoint M of AB. 1
- (ii) Show that the equation of the perpendicular bisector of AB is $3x + 2y - 5 = 0$. 3
- (iii) Find the coordinates of the point C that lies on the x-axis and is equidistant from A and B. 2
- (iv) The point D lies on the intersection of the line $y = 1$ and $3x + 2y - 5 = 0$. Find the coordinates of D and mark the position of D on your diagram. 3
- (b) Find the perpendicular distance of the point (-1, -2) from the line $6x + 5y = 30$. 3
- (c) Show that the lines $2y - 3x = 6$ and $6x + 9y = 14$ are perpendicular. 3
- (d) Find the equation of the line which passes through the point of intersection of $2y - 3x = 6$ and $4x + 6y = 5$, which is perpendicular to the line $3x - 5y = 15$. 5

Question 6 (23 marks)

(a) Differentiate:

(i) $4x^3 - 6x^2 + 3$ 1

(ii) $x - \frac{1}{4x^2}$ 2

(iii) $(\sqrt{x} + 1)^2$ 3

(iv) $\frac{x}{x^2 + 3}$ 2

(v) $(x-1)(x^2 + x + 1)$ 2

(b) Evaluate:

(i) $\lim_{x \rightarrow 1} \frac{x-1}{x^2 + x - 2}$ 3

(ii) $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{2x^2 - 3x + 7}$ 2

(c) Differentiate by first principles $y = x^2 + 4x - 6$. 4

(d) Show that the graph of $y = x^2 + 4x - 12$ crosses the x axis at two points.
Find the equations of the normals at these points. 4

Question 7 (24 marks)

(a) If α and β are the roots of $2x^2 + 5x - 1 = 0$ find the values of:

(i) $\alpha + \beta$ 1

(ii) $\alpha\beta$ 1

(iii) $\alpha^2 + \beta^2$ 2

(iv) $(\alpha + 1)(\beta + 1)$ 2

(v) $\frac{1}{\alpha} + \frac{1}{\beta}$ 2

(b) For the equation $x^2 + kx + (2k - 3) = 0$:

(i) find the two values of k for which the roots are equal 3

(ii) solve the equation for each of these values of k . 2

(c) Solve:

(i) $x^4 - x^2 - 12 = 0$ 4

(ii) $4^{2x} - 5 \times 4^x + 4 = 0$ 3

(d) Find the values of k for which $kx^2 + (k + 3)x + 4$ is positive definite and explain why there are no values of k for which it is negative definite. 4

$$Q1 a) \sqrt[3]{\frac{64 \cdot 9 \times 13 \cdot 4}{862 \times 3 \cdot 69}} = 0.65.$$

$$\begin{aligned} b) \frac{x-6}{x^2+4} + \frac{x+2}{x^2-4} \\ = \frac{x-6}{x^2+4} + \frac{x+2}{(x-2)(x+2)} \\ = \frac{(x-6)(x-2) + (x^2+4)}{(x^2+4)(x-2)} \\ = \frac{x^2-8x+12+x^2+4}{(x^2+4)(x-2)} \\ = \frac{2x^2-8x+16}{(x^2+4)(x-2)} \end{aligned}$$

$$\begin{aligned} c) i) 4x^2+16x-9 \\ = (4x^2-2x) + (18x-9) \\ = 2x(2x-1) + 9(2x-1) \\ = (2x+9)(2x-1) \end{aligned}$$

$$\begin{aligned} ii) 27-y^3 \\ = (3-y)(9+3y+y^2) \end{aligned}$$

$$\begin{aligned} d) \frac{1}{3-\sqrt{2}} + \frac{1}{3+\sqrt{2}} \\ = \frac{3+\sqrt{2} + 3-\sqrt{2}}{7} \\ = \frac{6}{7} \end{aligned}$$

$$\begin{aligned} e) 6x+3y &= 0 \quad (1) \\ 2x-5y &= 6 \quad (2) \times 3 \end{aligned}$$

$$\begin{aligned} 6x+3y &= 0 \\ 6x-15y &= 18 \end{aligned}$$

$$-18y = -18$$

$$y = -1$$

Sub $y = -1$ into (1)

$$6x + 3(-1) = 0$$

$$6x - 3 = 0$$

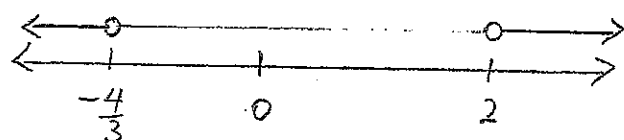
$$6x = 3$$

$$x = \frac{1}{2}$$

$$\therefore x = \frac{1}{2}, y = -1.$$

$$f) \text{ Solve } |3x-1| > 5$$

$$\begin{aligned} 3x-1 > 5 \quad \text{or} \quad 3x-1 < -5 \\ 3x > 6 \quad \quad \quad 3x < -4 \\ x > 2 \quad \quad \quad x < -\frac{4}{3} \end{aligned}$$



QUESTION 2.

$$\begin{aligned} a) i) \cos 150^\circ &= -\cos 30^\circ \\ &= -\frac{\sqrt{3}}{2} \end{aligned}$$

$$\begin{aligned} ii) \tan 300^\circ &= -\tan 60^\circ \\ &= -\sqrt{3} \end{aligned}$$

$$\begin{aligned} b) i) t^2 &= 7^2 + 9^2 - 2(7)(9)\cos 47^\circ \\ t^2 &= 44.068 \end{aligned}$$

$$\therefore t = 6.6 \text{ cm}$$

$$\begin{aligned} ii) A &= \frac{1}{2}(7)(9)\sin 47^\circ \\ &= 23.0 \text{ cm}^2 \end{aligned}$$

$$c) i) 2\cos \theta + 1 = 0$$

$$\begin{aligned} 2\cos \theta &= -1 \\ \cos \theta &= -\frac{1}{2} \end{aligned}$$

$$\therefore \theta = 120^\circ, 240^\circ$$

cii) $\tan^2 \theta = 3$

$\tan \theta = \pm \sqrt{3}$

$\theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ$

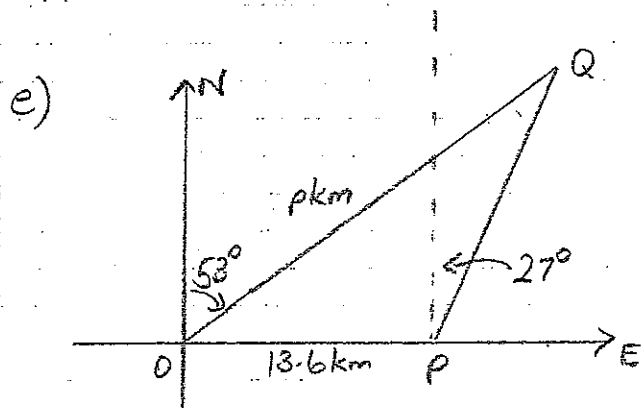
d) $\frac{x}{\sin 30^\circ} = \frac{25}{\sin 40^\circ}$

$\therefore x = \frac{25 \sin 30^\circ}{\sin 40^\circ}$

$\sin 70^\circ = \frac{h}{\frac{25 \sin 30^\circ}{\sin 40^\circ}}$

$h = \frac{25 \sin 30^\circ}{\sin 40^\circ} \times \sin 70^\circ$

$h = 18.3 \text{ m}$



$\angle QPO = 27^\circ + 90^\circ = 117^\circ$

$\angle QOP = 90^\circ - 53^\circ = 37^\circ$

$\angle OQP = 180^\circ - 117^\circ - 37^\circ = 26^\circ$

$\frac{p}{\sin 117^\circ} = \frac{13.6}{\sin 26^\circ}$

$p = \frac{13.6 \times \sin 117^\circ}{\sin 26^\circ}$

$p = 27.6 \text{ km}$

f) $\frac{1 - \cos^2 \theta}{1 - \sin^2 \theta}$

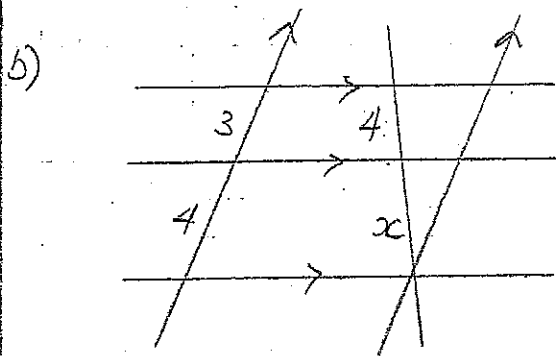
$= \frac{\sin^2 \theta}{\cos^2 \theta}$

$= \tan^2 \theta$

QUESTION 3.

a) $x = \frac{360^\circ}{8} = 45^\circ$ (exterior angle of n sided regular polygon)

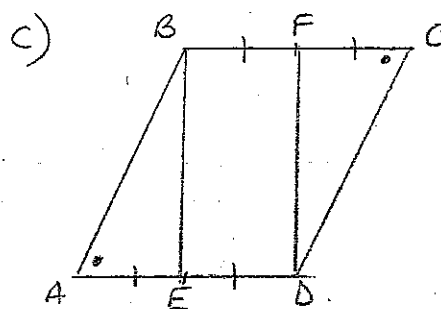
$y = \frac{180(n-2)^\circ}{n}$
 $= \frac{180 \times 6^\circ}{8} = 135^\circ$ (Interior angle of n sided regular polygon)



$\frac{x}{4} = \frac{4}{3}$ (ratio of intercepts)

$3x = 16$

$x = 5\frac{1}{3}$



In $\triangle ABE$ and $\triangle CDE$

$AB = CD$ (sides of a rhombus equal)
 $AE = CE$ (midpoint of sides of a rhombus)

$\angle BAE = \angle DCE$ (Opposite angles of a rhombus)

$\therefore \triangle ABE \cong \triangle CDE$ (SAS)

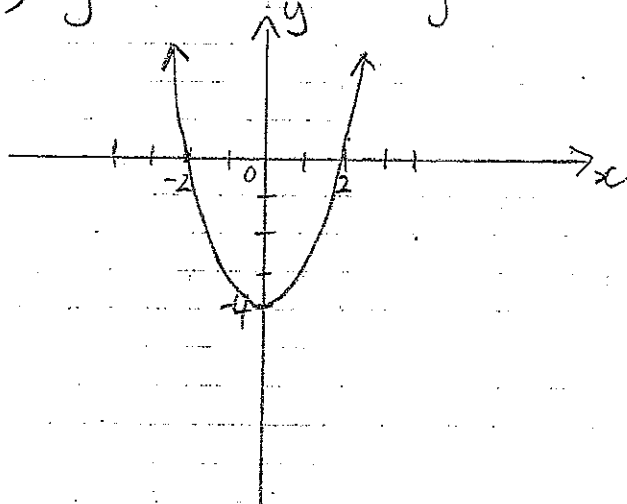
$BE = DE$ (midpoint of sides of rhombus)

$BE = DE$ (matching sides in congruent $\triangle$ s $\triangle ABE$ and $\triangle CDE$)

$\therefore EBFD$ is a parallelogram.

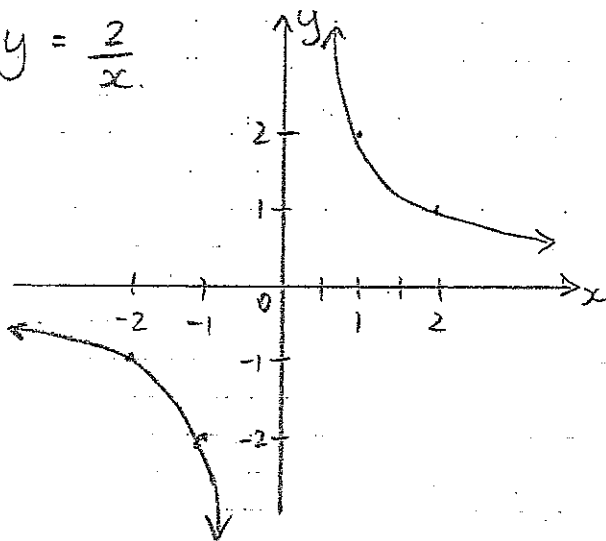
QUESTION 4

a) i) $y = x^2 - 4$ $y=0 \quad x=\pm 2$



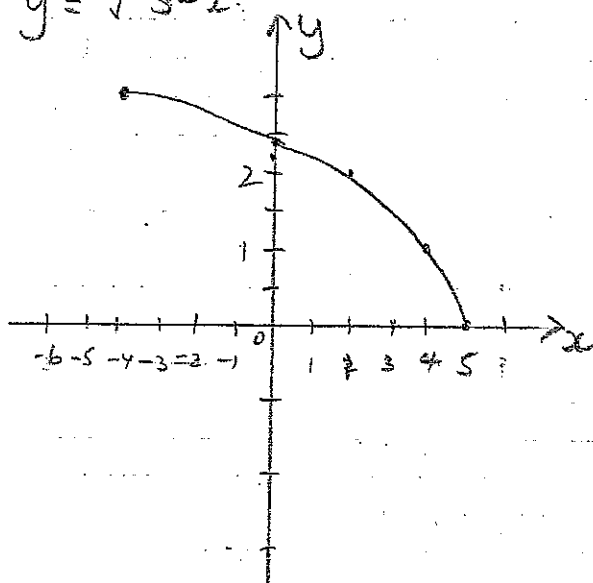
Domain: all real x .
Range: $y \geq -4$.

ii) $y = \frac{2}{x}$



Domain: all $x \quad x \neq 0$.
Range: all $y \quad y \neq 0$.

iii) $y = \sqrt{5-x}$



Domain: $x \leq 5$.
Range: $y \geq 0$.

b) $f(x) = x^3 + 4x$

$$f(-x) = (-x)^3 + 4(-x)$$

$$= -x^3 - 4x$$

$$= -(x^3 + 4x) = -f(x)$$

$$\therefore f(x) = -f(x)$$

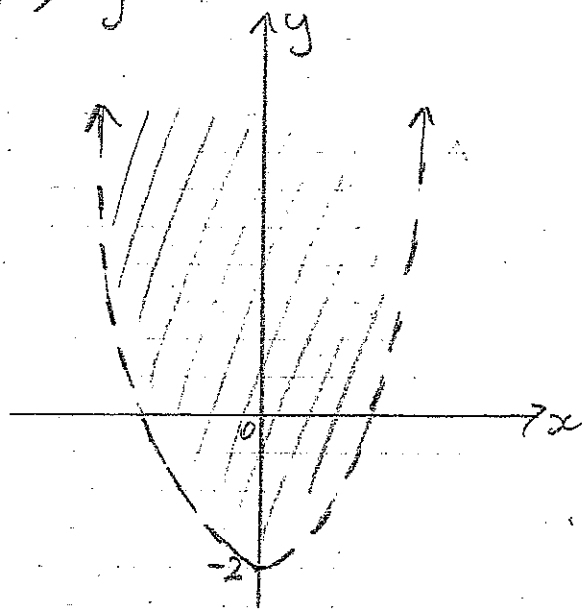
$\therefore$ The function is odd.

$$\begin{aligned} f(2) &= 2^3 + 4(2) \\ &= 8 + 8 \\ &= 16 \end{aligned}$$

$$\begin{aligned} f(-2) &= (-2)^3 + 4(-2) \\ &= -8 - 8 \\ &= -16 \end{aligned}$$

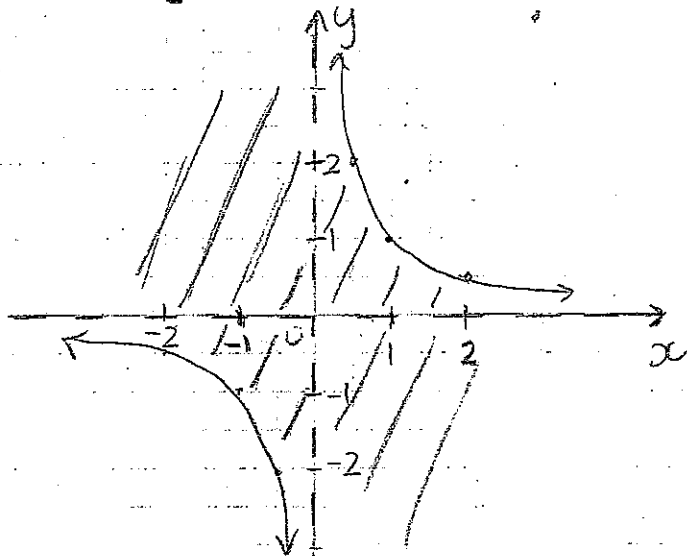
$$\therefore f(2) + f(-2) = 16 - 16 = 0$$

c) i) $y > x^2 - 2$



Test (0,0) $0 > 0^2 - 2$
 $0 > -2$

cii) $xy \leq 1$



Using $y \leq \frac{1}{x}$

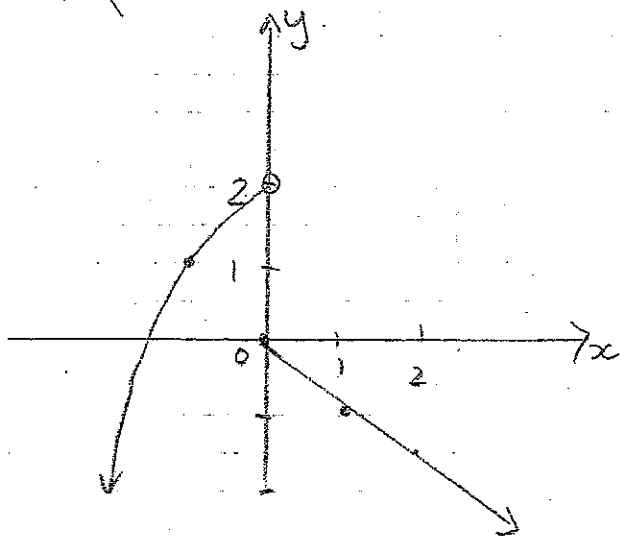
Test $(0,0)$ $0 \leq \frac{1}{0}$ undefined

Using $xy \leq 1$

Test $(2,1)$ $2 \leq 1$ not satisfied

Test $(-2,-2)$ $4 \leq 1$ not satisfied

d) $y = \begin{cases} 2-x^2 & x < 0 \\ -x & x \geq 0 \end{cases}$



QUESTION 5

ai) $m_{AB} = \left(\frac{-4+2}{-2}, \frac{2+6}{2} \right)$
 $= \left(\frac{-2}{-2}, \frac{8}{2} \right) = (-1, 4)$

ii) $m_1 = \frac{6-2}{2+4} = \frac{4}{6}$

$m_2 = -\frac{6}{4} = -\frac{3}{2}$

Equation of the perpendicular bisector of AB

$y-4 = -\frac{3}{2}(x+1)$

$y-4 = -\frac{3}{2}x - \frac{3}{2}$

$2y-8 = -3x-3$

$\therefore 3x+2y-5=0$

iii) C lies on the x axis.

so $y=0$

$3x+2(0)-5=0$

$3x=5$

$x = \frac{5}{3}$

$\therefore C\left(\frac{5}{3}, 0\right)$

iv) $y=1$ $3x+2y-5=0$

$3x+2-5=0$

$3x-3=0$

$3x=3$

$x=1$

Sub $x=1$ into $3x+2y-5=0$

$3+2y-5=0$

$2y-2=0$

$2y=2$

$y=1$

$\therefore D(1, 1)$

$$b) (-1, -2) \quad 6x + 5y = 30$$

$$d = \frac{|-6 - 10 - 30|}{\sqrt{6^2 + 5^2}}$$

$$= \frac{|-46|}{\sqrt{36+25}} = \frac{46}{\sqrt{61}}$$

$$c) 2y - 3x = 6, \quad 6x + 9y = 14$$

$$2y = 3x + 6$$

$$9y = -6x + 14$$

$$y = \frac{3}{2}x + 3$$

$$y = -\frac{2}{3}x + \frac{14}{9}$$

$$m_1 = \frac{3}{2}, \quad m_2 = -\frac{2}{3}$$

$$\therefore m_1 \times m_2 = \frac{3}{2} \times -\frac{2}{3} = -1$$

$\therefore 2y - 3x = 6$ is perpendicular to $6x + 9y = 14$

$$d) 2y - 3x = 6, \quad 4x + 6y = 5$$

$$6y + 4x = 5 \quad (1)$$

$$2y - 3x = 6 \quad (2) \times 3$$

$$6y + 4x = 5$$

$$6y - 9x = 18$$

$$13x = -13$$

$$x = -1 \quad \text{sub in (1)}$$

$$6y - 4 = 5$$

$$6y = 9$$

$$y = \frac{3}{2}$$

$\therefore$ Point of intersection $(-1, \frac{3}{2})$

Perpendicular gradient of

$$3x - 5y = 15$$

$$-5y = -3x + 15$$

$$y = \frac{3}{5}x - 3$$

$$m_1 = \frac{3}{5}, \quad m_2 = -\frac{5}{3}$$

Point gradient form

$$y - \frac{3}{2} = -\frac{5}{3}(x + 1)$$

$$y - \frac{3}{2} = -\frac{5}{3}x - \frac{5}{3}$$

$$3y - \frac{9}{2} = -5x - 5$$

$$6y - 9 = -10x - 10$$

$$\therefore 10x + 6y + 1 = 0$$

$\therefore$ The equation of the line is $10x + 6y + 1 = 0$

QUESTION 6

$$a) i) 4x^3 - 6x^2 + 3$$

$$\frac{dy}{dx} = 12x^2 - 12x$$

$$ii) x - \frac{1}{4x^2} = x - \frac{1}{4}x^{-2}$$

$$\frac{dy}{dx} = 1 + \frac{1}{2}x^{-3}$$

$$= 1 + \frac{1}{2x^3}$$

$$iii) (\sqrt{x} + 1)^2$$

$$\frac{dy}{dx} = 2(\sqrt{x} + 1) \times \frac{1}{2\sqrt{x}}$$

$$= \frac{2(\sqrt{x} + 1)}{2\sqrt{x}}$$

$$= \frac{\sqrt{x} + 1}{\sqrt{x}}$$

$$iv) \frac{x}{x^2+3}$$

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$$

$$u = x \quad u' = 1$$

$$v = x^2+3 \quad v' = 2x$$

$$\frac{dy}{dx} = \frac{(x^2+3) \cdot 1 - x(2x)}{(x^2+3)^2}$$

$$= \frac{x^2+3-2x^2}{(x^2+3)^2}$$

$$= \frac{3-x^2}{(x^2+3)^2}$$

$$v) (x-1)(x^2+x+1)$$

$$\frac{dy}{dx} = vu' + uv'$$

$$u = x-1 \quad u' = 1$$

$$v = x^2+x+1 \quad v' = 2x+1$$

$$\frac{dy}{dx} = (x^2+x+1) \cdot 1 + (x-1)(2x+1)$$

$$= (x^2+x+1) + (2x^2-x-1)$$

$$= x^2+x+1+2x^2-x-1$$

$$= 3x^2$$

$$bi) \lim_{x \rightarrow 1} \frac{x-1}{x^2+x-2}$$

$$\lim_{x \rightarrow 1} \frac{x-1}{(x-1)(x+2)}$$

$$\lim_{x \rightarrow 1} \frac{1}{x+2}$$

$$= \frac{1}{3}$$

$$ii) \lim_{x \rightarrow \infty} \frac{3x^2+2x}{2x^2-3x+7}$$

$$\lim_{x \rightarrow \infty} \frac{\frac{3x^2}{x^2} + \frac{2x}{x^2}}{\frac{2x^2}{x^2} - \frac{3x}{x^2} + \frac{7}{x^2}}$$

$$\lim_{x \rightarrow \infty} \frac{3 + \frac{2}{x}}{2 - \frac{3}{x} + \frac{7}{x^2}}$$

$$= \frac{3}{2}$$

$$c) y = x^2+4x-6$$

$$f(x) = x^2+4x-6$$

$$f(x+h) = (x+h)^2+4(x+h)-6$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2+4(x+h)-6 - (x^2+4x-6)}{h}$$

$$\lim_{h \rightarrow 0} \frac{x^2+2xh+h^2+4x+4h-6 - x^2-4x+6}{h}$$

$$\lim_{h \rightarrow 0} \frac{2xh+h^2+4h}{h}$$

$$\lim_{h \rightarrow 0} 2x+h+4$$

$$= 2x+4$$

$$d) y = x^2 + 4x - 12$$

$$y=0 \quad x^2 + 4x - 12 = 0 \\ (x+6)(x-2) = 0$$

$$x = -6 \text{ or } x = 2.$$

$$\frac{dy}{dx} = 2x + 4 \quad \text{When } x = -6 \quad \frac{dy}{dx} = -8 \\ \text{When } x = 2 \quad \frac{dy}{dx} = 8$$

$$\text{At } (-6, 0) \text{ Gradient of normal} = \frac{1}{8}$$

Equation of normal at $(-6, 0)$

$$y - 0 = \frac{1}{8}(x + 6) \\ y = \frac{x}{8} + \frac{6}{8} \text{ or } x - 8y + 6 = 0$$

$$\text{At } (2, 0) \text{ Gradient of normal} = -\frac{1}{8}$$

Equation of normal at $(2, 0)$

$$y - 0 = -\frac{1}{8}(x - 2) \\ y = -\frac{x}{8} + \frac{1}{4} \text{ or } x + 8y - 2 = 0$$

QUESTION 7

$$a) 2x^2 + 5x - 1 = 0$$

$$a = 2 \quad b = 5 \quad c = -1$$

$$i) \alpha + \beta = \frac{-b}{a} = \frac{-5}{2}$$

$$ii) \alpha\beta = \frac{c}{a} = \frac{-1}{2}$$

$$iii) \alpha^2 + \beta^2 = (\alpha^2 + 2\alpha\beta + \beta^2) - 2\alpha\beta \\ = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= \left(\frac{-5}{2}\right)^2 - 2\left(\frac{-1}{2}\right)$$

$$= \frac{25}{4} + 1 = \frac{29}{4}$$

$$iv) (\alpha + 1)(\beta + 1)$$

$$= \alpha\beta + \alpha + \beta + 1$$

$$= \alpha\beta + (\alpha + \beta) + 1$$

$$= \frac{-1}{2} + \left(\frac{-5}{2}\right) + 1$$

$$= \frac{-1}{2} - \frac{5}{2} + 1$$

$$= -3 + 1 = -2$$

$$v) \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta + \alpha}{\alpha\beta}$$

$$= \frac{-\frac{5}{2}}{\frac{-1}{2}} = -\frac{5}{2} \times \frac{-2}{1} = \frac{10}{2} = 5$$

$$b) x^2 + kx + (2k - 3) = 0$$

$$i) \Delta = 0 \quad b^2 - 4ac = 0 \\ a = 1 \quad b = k \quad c = (2k - 3) \\ \Delta = k^2 - 4(1)(2k - 3)$$

$$= k^2 - 4(2k - 3) \\ = k^2 - 8k + 12$$

$$\text{When } k^2 - 8k + 12 = 0$$

$$(k - 6)(k - 2) = 0$$

$$\therefore k = 6 \text{ or } k = 2$$

$$ii) \text{When } k = 6$$

$$x^2 + 6x + 9 = 0$$

$$(x + 3)^2 = 0$$

$$\therefore x = -3$$

$$\text{When } k = 2$$

$$x^2 + 2x + 1 = 0$$

$$(x + 1)^2 = 0$$

$$\therefore x = -1$$

$$c) i) x^4 - x^2 - 12 = 0$$

$$\text{Let } u = x^2$$

$$u^2 - u - 12 = 0$$

$$(u - 4)(u + 3) = 0$$

$$u = 4 \text{ or } -3$$

$$\therefore x^2 = 4 \text{ or } x^2 = -3$$

$$\text{as } x^2 \geq 0 \quad x^2 = -3 \text{ has no real solution}$$

$$\therefore x = \pm 2 \text{ is the only solution}$$

$$\text{ii) } 4^{2x} - 5 \times 4^x + 4 = 0$$

$$\text{Let } u = 4^x$$

$$u^2 - 5u + 4 = 0$$

$$(u-4)(u-1) = 0$$

$$\therefore u = 4 \text{ and } 1$$

$$4^x = 4 \text{ and } 4^x = 1$$

$$\therefore x = 1 \text{ and } 0$$

$$\text{d) } kx^2 + (k+3)x + 4$$

Positive definite $a > 0$, $\Delta < 0$.

$$\therefore b^2 - 4ac < 0$$

$$(k+3)^2 - 4(k)(4) < 0$$

$$(k+3)^2 - 16k < 0$$

$$k^2 + 6k + 9 - 16k < 0$$

$$k^2 - 10k + 9 < 0$$

$$(k-9)(k-1) < 0$$

$$1 < k < 9$$

There are no values of k for which it is negative definite as all values of k are positive therefore $a > 0$.